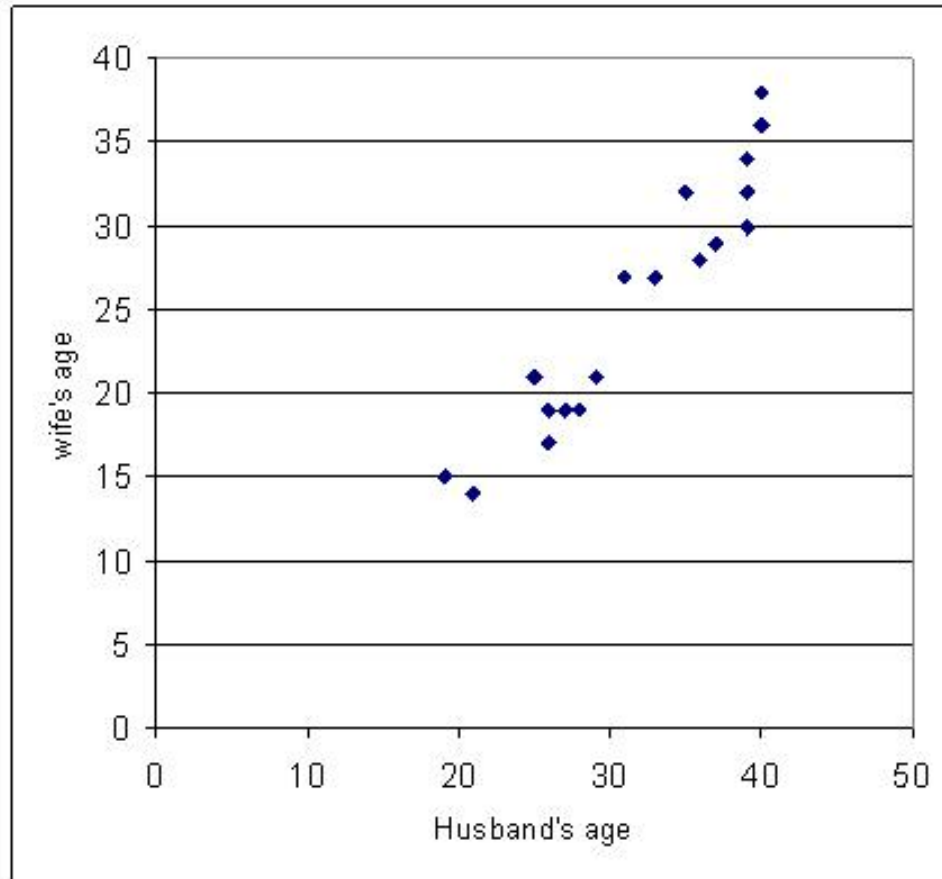


Scatter Plot

- A set of bivariate (data with two variables) graphed as ordered pairs in a coordinate plane is called a scatter plot.
- A **scatter plot** shows relationships between two sets of data.

You don't connect
the points in a
scatter plot.

Scatter Diagram



Scatter Plot

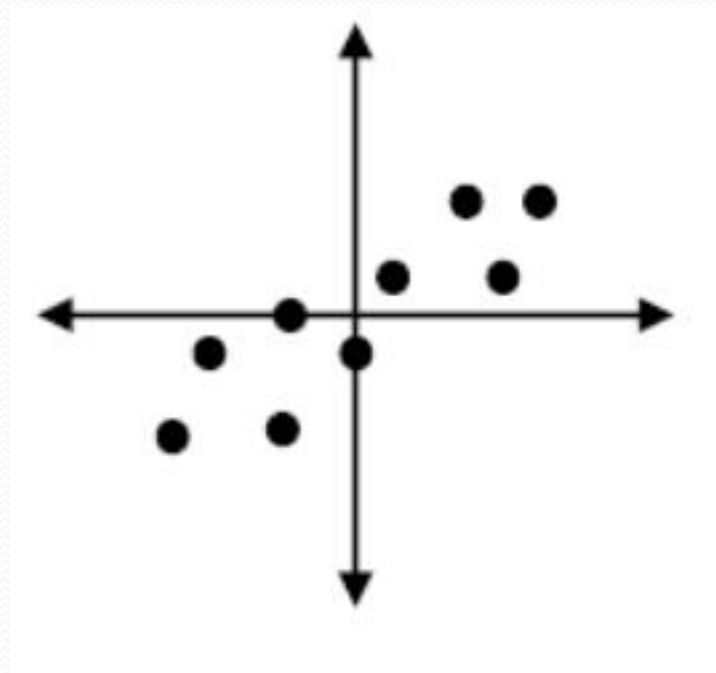
- Scatter plots are similar to line graphs in that they use horizontal and vertical axes to plot data points.
- However, they have a very specific purpose. Scatter plots show how much one variable is affected by another.
- The relationship between two variables is called their **correlation**.

Correlation

- The relationship between two variables is called their **correlation** .

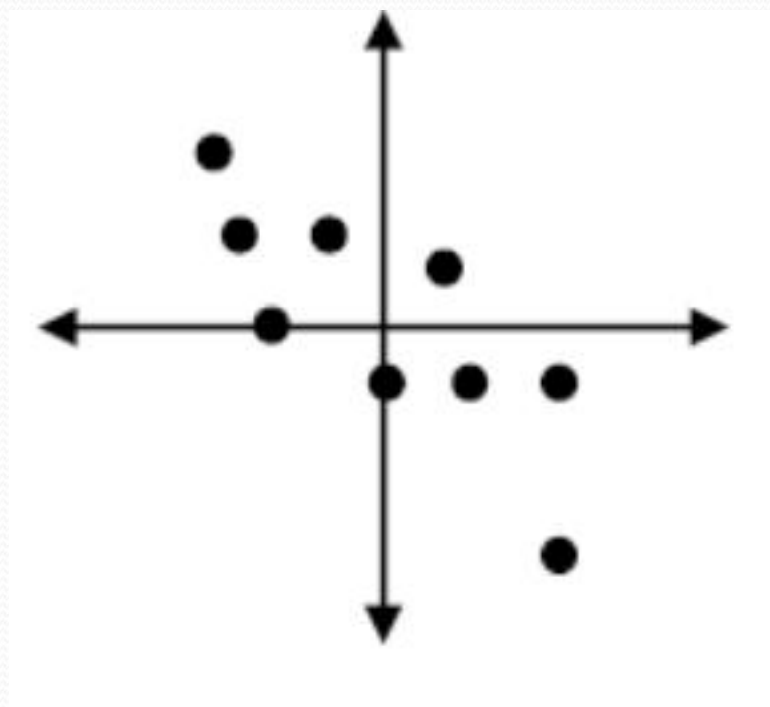
Positive Correlation

- If the data points make a straight line going from the origin out to high x- and y-values, then the variables are said to have a **positive correlation**



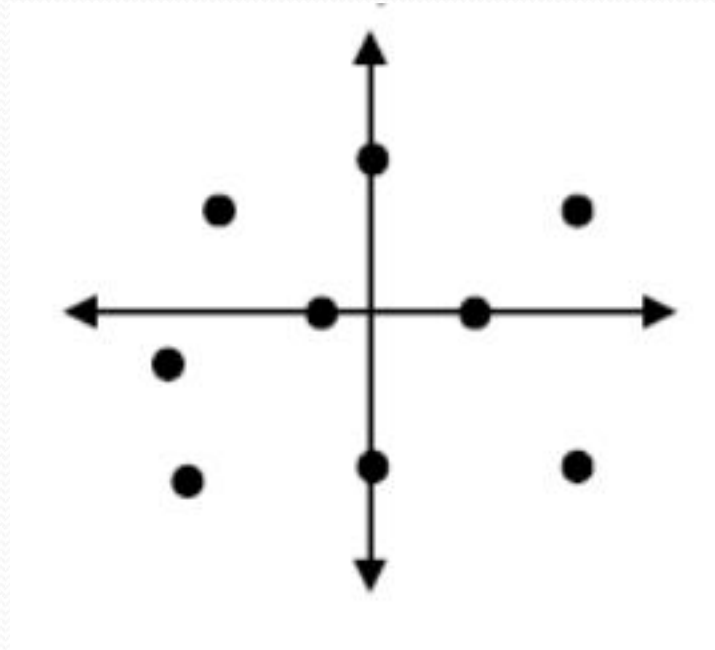
Negative Correlation

- If the line goes from a high-value on the y-axis down to a high-value on the x-axis, the variables have a **negative correlation**.



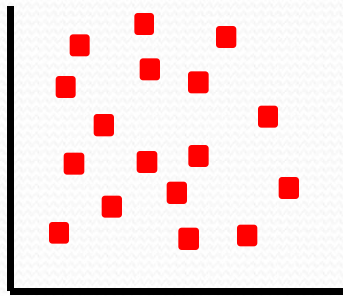
No Correlation

- If the plot on the graph is scattered in such a way that it does not approximate a line (it does not appear to rise or fall), there is **no correlation** between the sets of data.

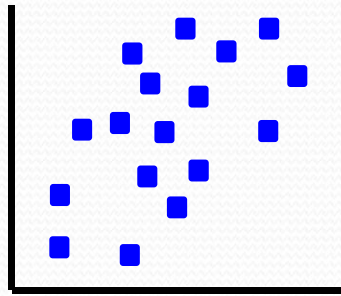


Scatter Diagram - How do I use it? - Correlation

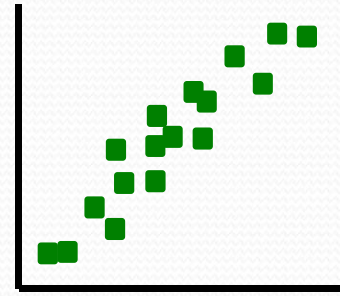
Degrees of correlation:



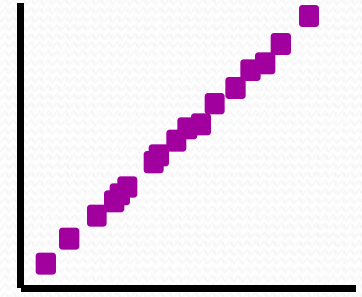
None



Low

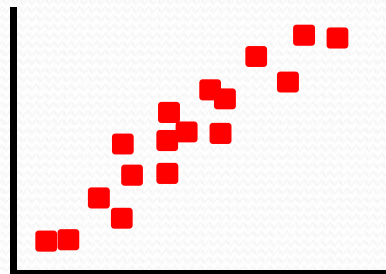


High

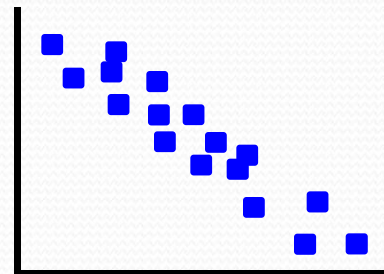


Perfect

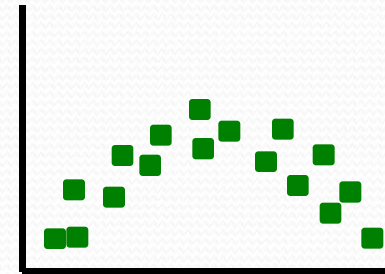
Types of correlation:



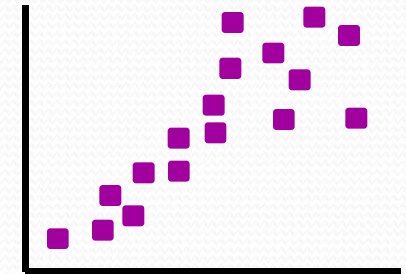
Positive



Negative



Curved

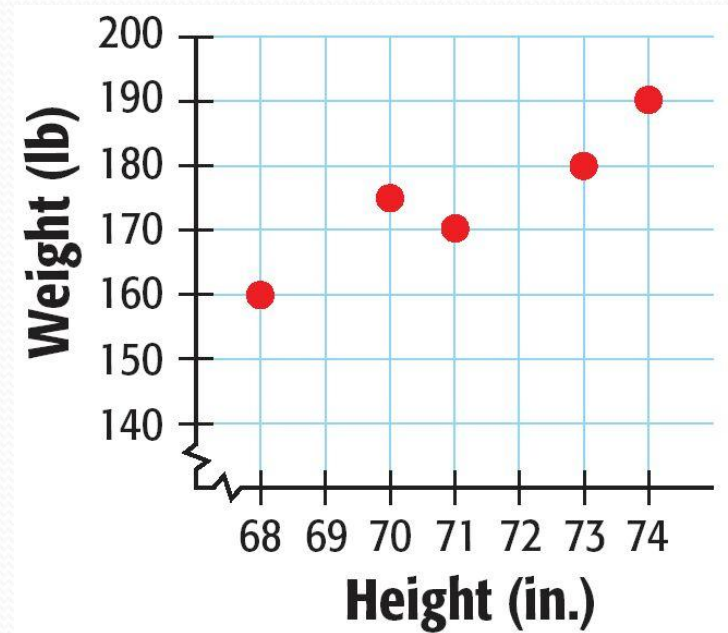


Partial

Example

- Use the given data to make a scatter plot of the weight and height of each member of a basketball team.

Height (in.)	Weight (lb)
71	170
68	160
70	175
73	180
74	190



Is there a relationship between the fat grams and the total calories in fast food?

Sandwich	Total Fat (g)	Total Calories
Hamburger	9	260
Cheeseburger	13	320
Quarter Pounder	21	420
Quarter Pounder with cheese	30	530
Big Mac	31	560
Sandwich special	31	550
Special with bacon	34	590
Crispy Chicken	25	500
Fish Fillet	28	560
Grilled Chicken	20	440
Grilled Chicken Light	5	300

Uses of Scatter Diagram:

- To understand the behaviour of a process.
- To determine if there is a relationship between two factors.
- To visually demonstrate the correlation between two related factors.
- To determine where there may be a cause and effect relationship.